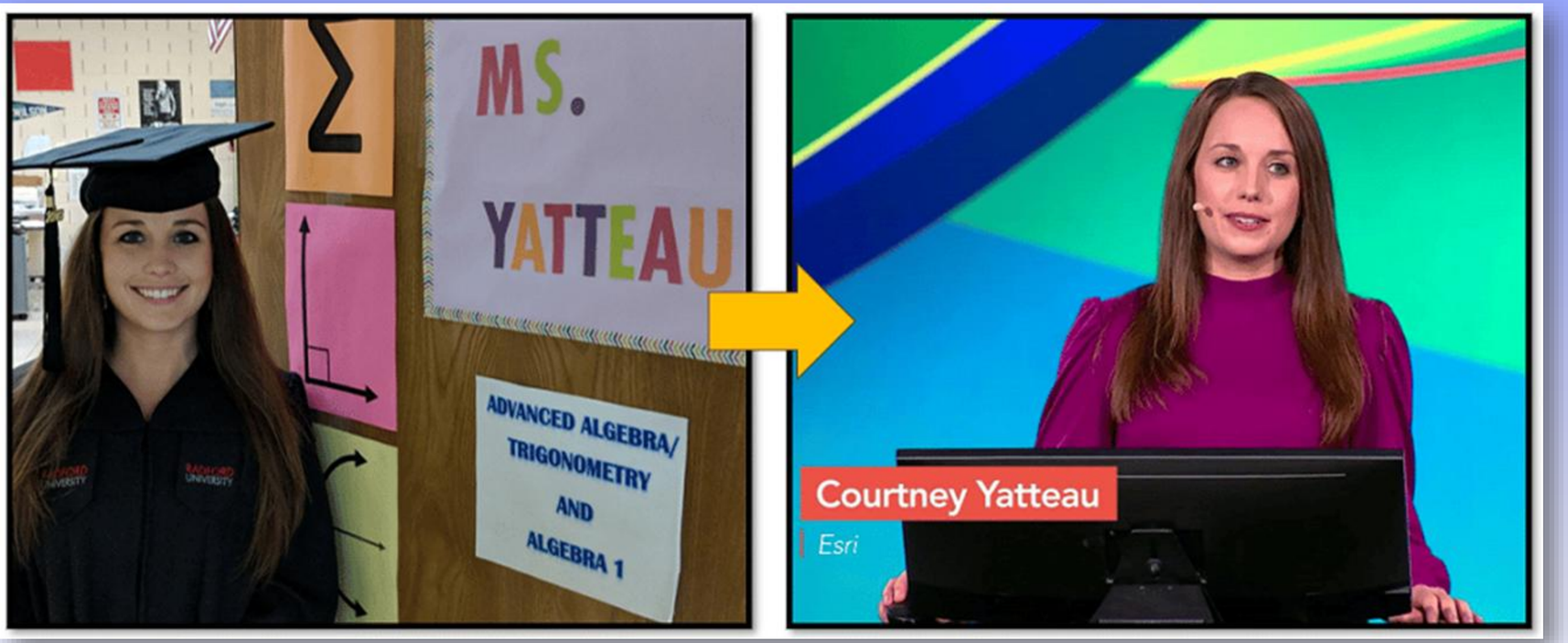
Courtney Yatteau

ABOUT ME



Courtney Yatteau

Developer Advocate, Esri

Agenda

1. Gamification basics
2. Three practical AI patterns
3. One combined app
4. How to build this in practice

Why Gamification Feels Shallow

Points feel meaningless

Badges no one cares about

Feedback feels generic

Difficulty feels random

Good Gamification

PROGRESS

FEEDBACK

CHALLENGE

REWARD

WHY NOW?

AI gamification is practical now

Hosted AI

Generate

Classify

Browser AI

Adapt locally

Fast feedback

Smaller AI loops

One useful signal

One useful change

Don't overbuild.

Micro-Glossary

01

SIGNAL

What the app observes

02

DECISION

What the AI does with that signal

03

RESPONSE

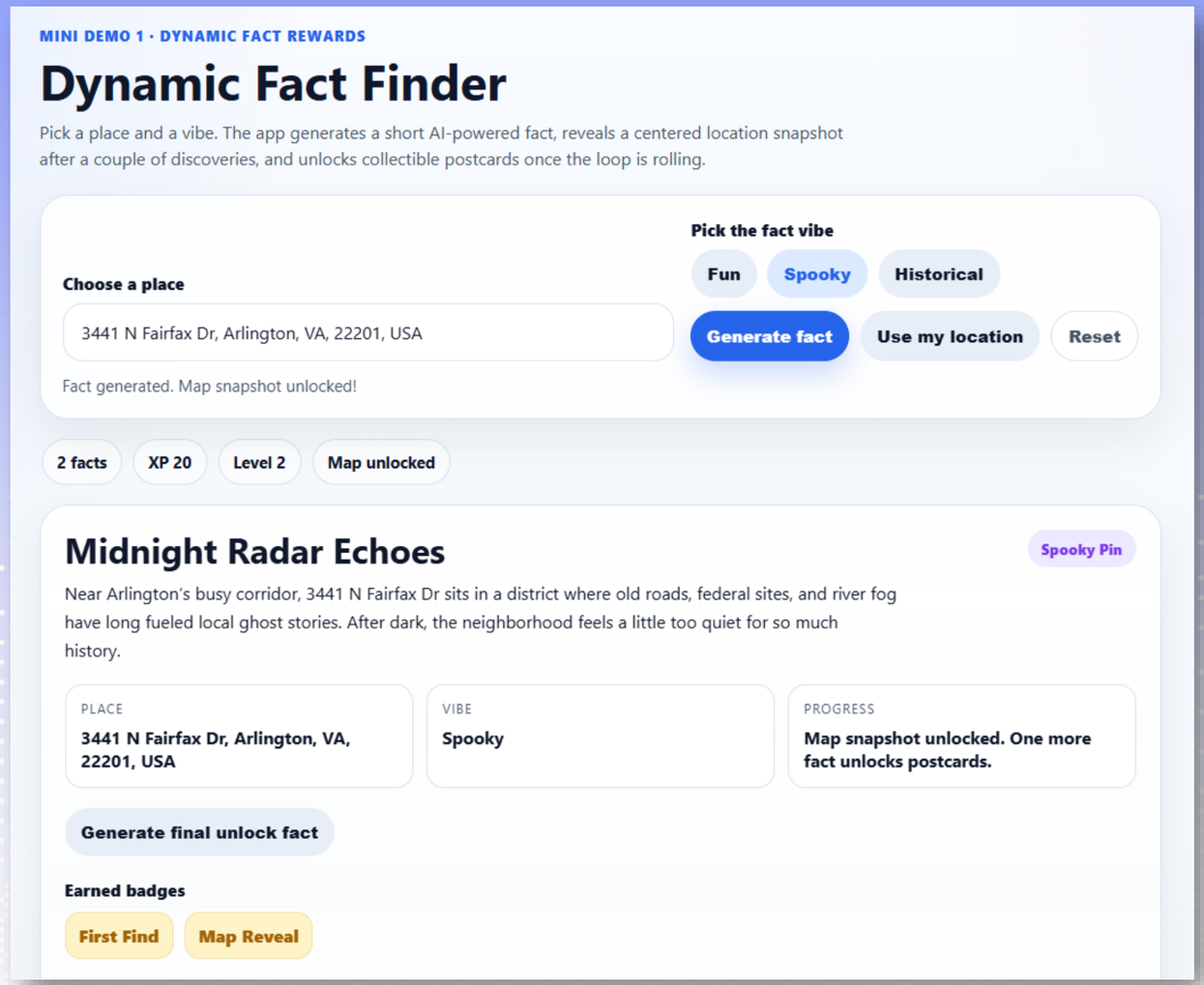
How the UI changes

Dynamic Content

Generate fresh prompts, facts,
or next-step variations
without a giant content system

VARIETY REPLAYABILITY LIGHTWEIGHT PERSONALIZATION

Mini Demo: Dynamic Fact Finder



SIGNAL

Click

Fact type

Location

DECISION

Generate a structured fact card

RESPONSE

Fresh content

Reward label

Progress update

Make AI output usable

```
format: {
  type: "json_schema",
  strict: true,
  schema: {
    properties: {
      headline: { type: "string" },
      fact: { type: "string" },
      rewardLabel: { type: "string" }
    },
    required: ["headline", "fact", "rewardLabel"]
  }
}
```

Schema-shaped output

Strict response contract

UI-friendly fields

What Powered this Demo

Place grounding

ArcGIS autosuggest

Place resolution

Reverse geocode

Content generation

OpenAI Responses API

Structured Outputs

GPT-5.4

Map reveal

Static Basemap Tiles

Centered snapshot crop

arcgis/community style

Reward

XP, badges, levels

GPT Image 1.5 postcard

Local unlock state

🔑 **Server-side keys**

🌐 **HTTPS for location**

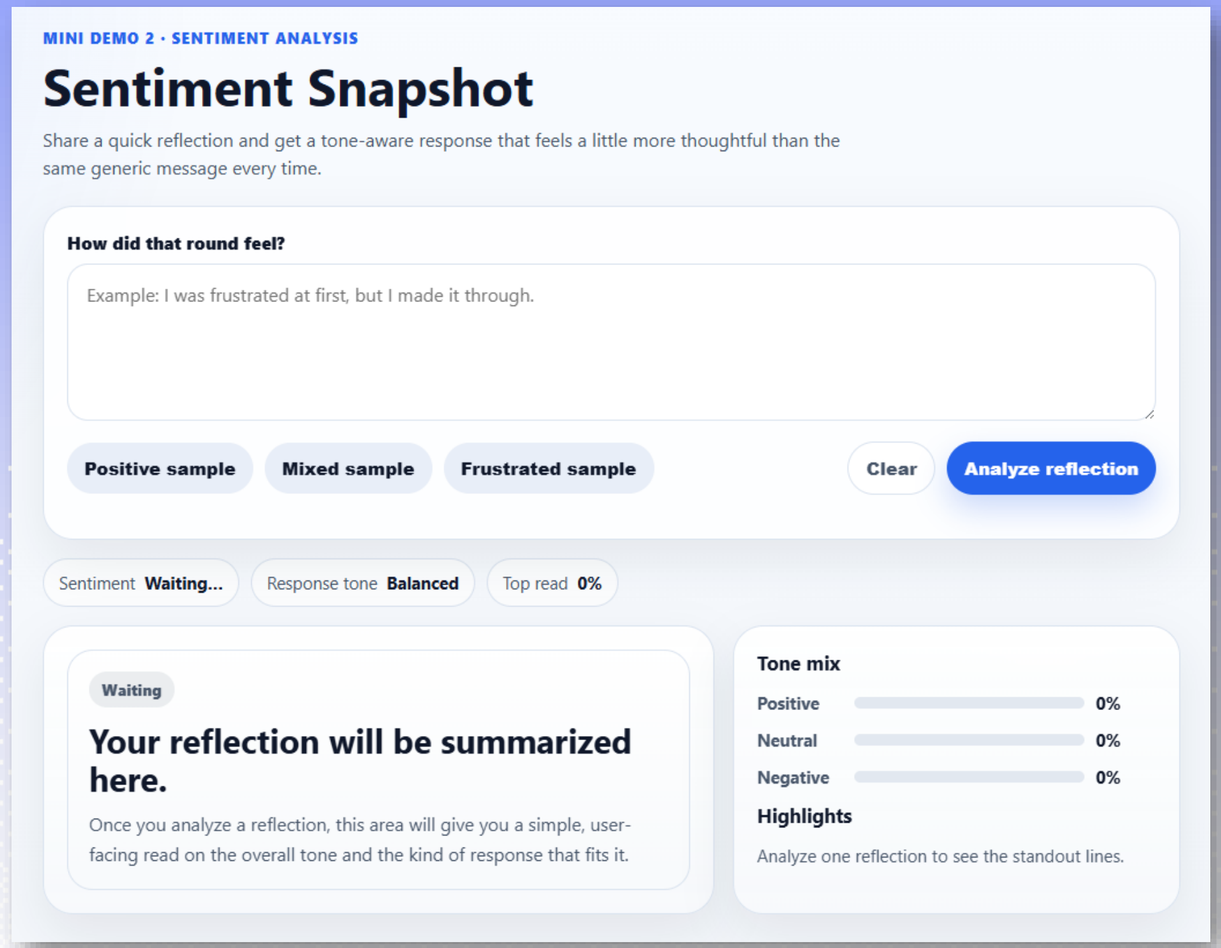
Sentiment-Aware Feedback

React to tone and
choose a better feedback path

Tone Detection Better Reinforcement Smarter Follow-Up

Post-round checkpoint

Mini Demo: Feedback that Reacts



SIGNAL

Player reflection

Sentence tone shifts

DECISION

Choose feedback route

Use confidence scores

RESPONSE

Tone changes

Recovery options change

Make AI output usable

```
const payload = {
  kind: "SentimentAnalysis",
  analysisInput: {
    documents: [{ id: "1", language: "en", text }]
  }
};
```

Analyze the reflection

```
const route = chooseRoute(doc);

if (route.key === "negative") {
  // protect progress, unlock support
} else if (route.key === "mixed") {
  // keep checkpoint, offer retry
}
```

Choose a route

Change the UI

What Powered this Demo

Player Input

Reflection after a round

Signal result

Document + sentence sentiment

Confidence scores

App response

Choose checkpoint route

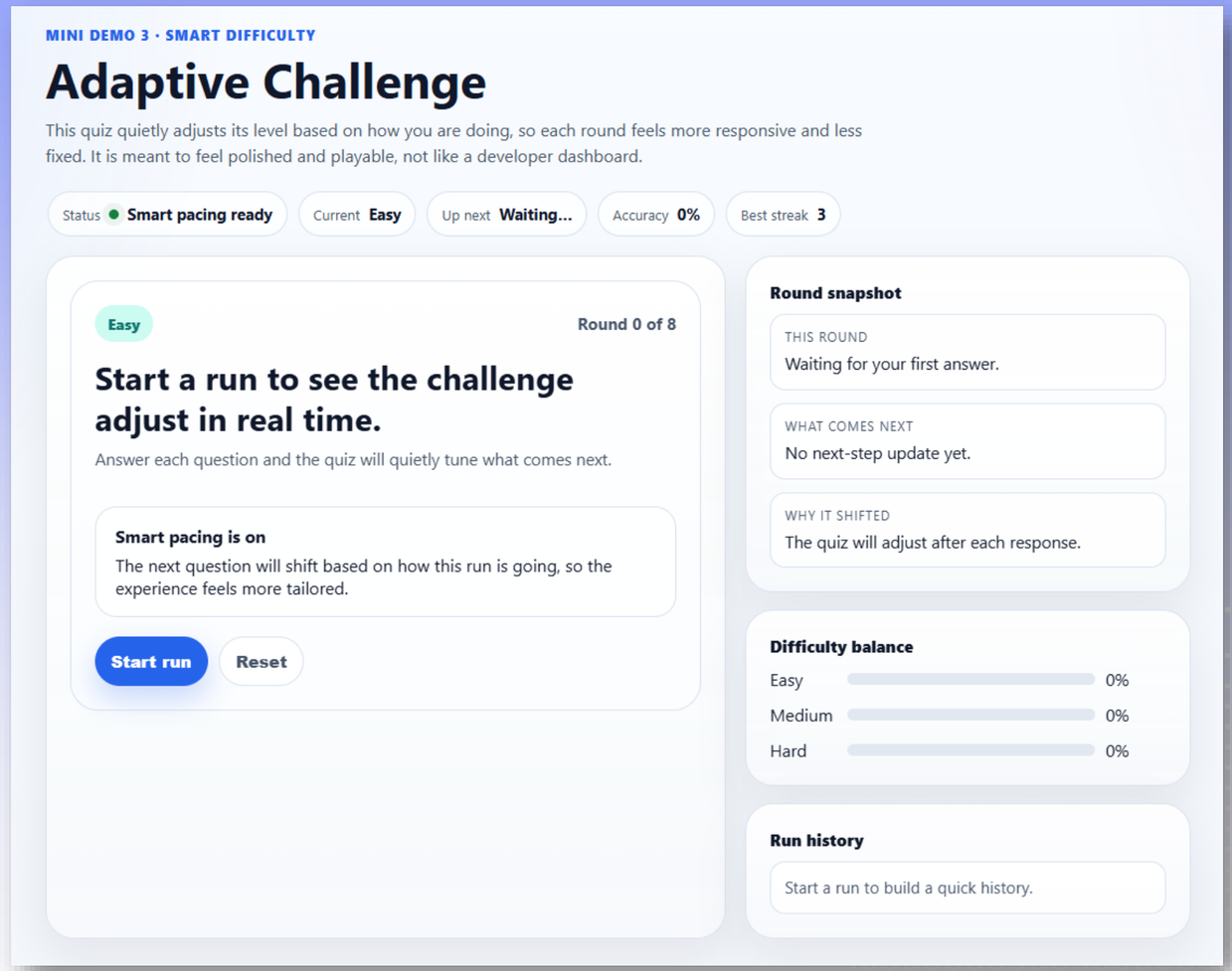
Tone, support, and recovery

Adaptive Challenge Tuning

React to correctness, timing,
and repeated behavior
to keep challenge balanced

Behavior Signals **Live Adjustment** **Better Pacing**

Mini Demo: Difficulty that Adapts



SIGNAL

Correctness

Response Time

DECISION

Next Prediction

RESPONSE

Current Difficulty

Make AI output usable

```
const features = [
  correct ? 1 : 0,
  responseTime,
  currentDifficulty
]
```

Capture features

Predict next state

```
const input = tf.tensor2d([features])
const prediction = model.predict(input)
currentDifficulty = prediction.argMax(-1).dataSync()[0]
```

```
difficultyLevel.textContent = ['Easy', 'Medium', 'Hard'][currentDifficulty]
setTimeout(nextQuestion, 2000)
```

Update UI

What Powered this Demo

Browser-side model

TensorFlow.js

Small local prediction loop

Inputs

Correctness

Response time

Current difficulty

Decision + UI Response

Predict next difficulty

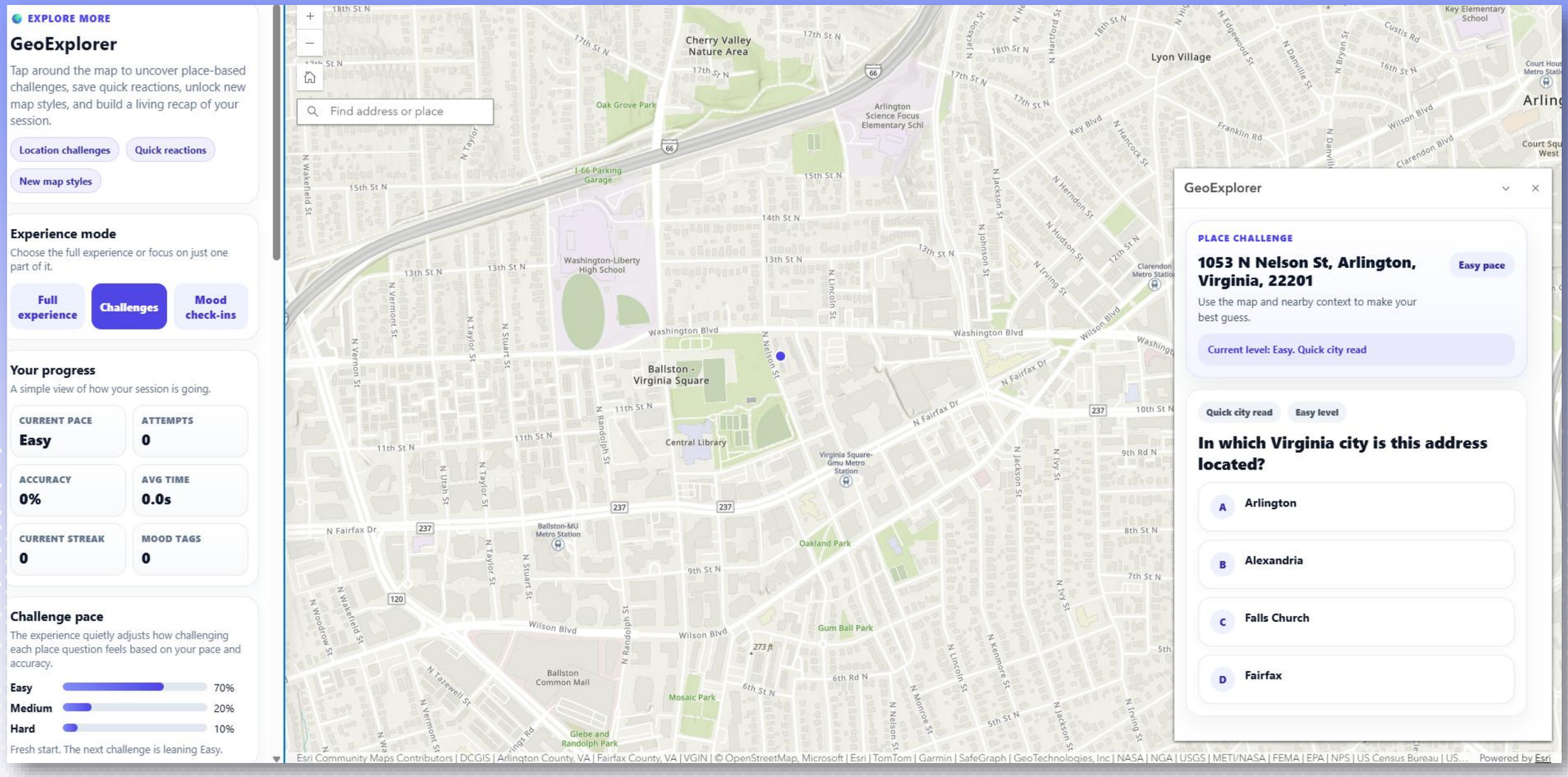
Change what comes next

Small model, noticeable pacing shift

The interesting part isn’t each feature.

It’s how the loops work together.

GeoExplorer



adaptive quiz mode **sentiment recap** **badges + unlocks**

The reusable loop

01	**Signals in** clicks, answers, timing, tone, or streaks
02	**AI decides** generate, classify, summarize, or adjust
03	**UI responds** prompt, reward, unlock, recap or retune

WHAT MAKES THIS WORK

- Better context
- Better feedback loops
- Visible UI change
- Useful, not magical

THANKS FOR PLAYING

Thank you, Frontrunners!

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github.com/cyatteau/ai-gamification-frontrunners2026

GAME OVER